

Fujitsu FRAM (Ferroelectric RAM) Product

FRAM technology cumulates SRAM's speed advantage and E2PROM's non-volatility into a single product. With the use of the ferroelectric film inside the FRAM, data is modified by the polarization of this film in the presence of an electric field, and the polarization state remains even when the electric field is removed.

Comparison between FRAM and other memories

	FRAM	EEPROM	FLASH	SRAM
Memory Type	Non-volatile	Non-volatile	Non-volatile	Volatile
Write Method	Overwrite	Erase + Write	Erase + Write	Overwrite
Write Cycle Time	150ns	5ms	10µs	55ns
Read/Write Cycles	10 ¹³	10 ⁶	10 ⁶	Unlimited
Booster Circuit	No	Yes	Yes	No
Data Backup Battery	No	No	No	Yes

Target Application

Vehicle Tracking/Management



1 & 3 Phase Energy meter



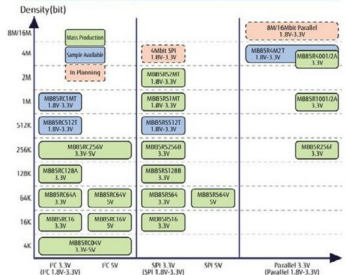
Office Automation Equipment/
Digital Analyzers



POS/ATM/ Medical,
Pharmaceutical tracking



FRAM Roadmap



FRAM is available with Fujitsu distributors who can sell in local Indian rupee & maintain sufficient stock

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